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MILITARY SPECIFICATION

HEATER, ENGINE AND SHELTER, GROUND

TYPE H-1

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel fuel or gasoline-burning, self-contained, ground portable heaters, designated type H-1, having a nominal uncontaminated warm air heat output of 400,000 Btu/hr. The heaters are operated as entirely self-sustained heaters by using either a diesel or gasoline engine power package, or as slave-type heaters using an electric motor power package.

1.2 Classification. The heater shall be of the following classes, as specified (see 6.2):

Class 1	- Equipped with a Commerical Gasoline Engine Power Package Conforming to MIL-STD-1410.
Class 2	- Equipped with an Electric Power Package.
Class 3	- Equipped with a Military Standard Gasoline Engine, 1-1/2 bhp, Model 1A08-III.
Class 4	- Equipped with a Diesel Engine Conforming to MIL-STD-141O.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GG-P-455	- Plate, Photographic (Photosensitive Aluminum) .
W-F-800	- Fuel Oil, Diesel.
w-G-1690	- Gasoline, Automotive, Leaded or Unleaded.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.

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MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-G-5572	- Gasoline, Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-5756	- Cable and Wire, Power, Electric, portable.
MIL-H-7365	- Hose, Air Duct, for Ground Heaters.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-W-8005	- Wheels arid Hubs, for Industrial Pneumatic Tires.
MIL-A-8421	- Air Transportability Requirements, General Specification for.
MIL-E-1OO62	- Engine, Preparation for Shipment and Storage of.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-T-22085	- Tape, Pressure Sensitive, Adhesive, Preservation and Sealing.

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MIL-C-46168 - Coating, Aliphatic, Polyurethane, Chemical Agent Resistant.
MIL-T-46786 - Tank, Fuel, Engine, General Requirements for.
MIL-T-83133 - Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-H28 Screw-Thread Standards for Federal Service.
FED-STD-101 - Preservation, Packaging and Packing Materials, Test Procedures.
FED-STD-595 - Colors.

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DOD-STD-1OO - Engineering Drawing Practices.
MIL-STD-105 Sampling Procedures and Tables for lnspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 Identification Marking of US Military Property.
MIL-STD-171 Finishing of Metal and Wood Surfaces.
*MIL-sTD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-81O - Environmental Test Methods.
MIL-STD-882 - System Safety Program Requirements.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, and Appropriate Test Methods.
MIL-STD-1226 - Engine, Gasoline, Air cooled) 1-1/2 BHP, - Cycle, Military Standard, Model 1A08, Installation Procedures.
M1L-STD-141O Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS 35000 - Battery, Storage, Lead-Acid, WaterProof.

* See paragraph 3.5.13 for the applicable issue of MIL-STD-461.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DEPARTMENT OF AGRICULTURE FOREST SERVICE

Standard 5100-1 - Standard for Spark Arrestors for Internal Combustion Engines.

(Application for copies should be addressed to the US Forest Service, San Dimas Equipment Development Center, 444 East Bonita Ave, San Dimas, CA 91773.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation, Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Power Test Code, 19.5:4, Supplement on Instruments and Apparatus; Part 5, Chapter 4, Flow Measurements by Means of Standardized Nozzles and Orifice Plates.

Power Test Code, Supplement on Instruments and Apparatus, Part 10, Flue Exhaust Gas Analysis.

Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AS 1055 - Fire Testing of Flexible Hose, Tube Assemblies, Coils, Fittings and Similar System Components.
J 429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J 492 - Rivets and Riveting.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification

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(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJune Road, P.O. Box 351040, Miami, FL 33135.)

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

AMCA Standard 210 - Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Movement and Control Association, 30 West University Drive, Arlington Heights, IL 60004.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS (ASHRAE)

ASHRAE Handbook and Product Directory - 1985 Fundamentals.
ASHRAE Standard 51 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standard MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The heater shall be as specified herein. The heater shall be of the liquid fuel burning, combustion type with heated, forced air delivered through flexible air ducts to the space and objects to be heated. The heater assembly shall be fully enclosed in a metal cabinet and shall be integrally-

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mounted on a two-wheeled chassis for local portability. The heater assembly shall have provisions for stowing of all accessories and all separable components.

3.1.1 Components. The heater shall consist of the following major components:

Quantity	Item or System Name	See Rqmt.
1	Burner Assembly	3.6.1.1
1	Combustion Chamber	3.6.1.2
1	Heat Exchanger	3.6.1.3
1	Air System	3.6.2
1	Fuel System	3.6.3
1	Ignition System	3.6.4
1	Heat Control System	3.6.5
1	Power Package System	3.6.6
1	Housing	3.6.7
As Required	Accessories	3.6.8

3.1.2 Size and weight. The overall size of the complete heater assembly with all accessories stowed for transit mode shall not exceed 42 inches height, 55 inches width, and 70 inches length with the tow bar positioned for parking, if applicable (see 3.6.7.7). The dry weight of the complete heater assembly with all accessories for class 1, 2, or 3 heaters shall not exceed 675 pounds. The dry weight of the class 4 heater assembly with all accessories shall not exceed 800 pounds.

3.2 First article. When specified in the contract, a sample shall be subjected to first article inspection (see 4.3 and 6.3), to prove prior to starting production that his production methods and choice of design detail will produce heaters that comply with the requirements of this specification. The inspection as specified in section 4 shall be conducted by the contractor subject to surveillance and approval by the Government.

3.3 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

3.3.1 Material deterioration prevention and control. The heaters shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the heater may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Protective treatment. When materials are used in the construction of the heater that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip or scale with age or extremes at atmospheric conditions shall be avoided.

3.3.5 Material safety. Materials selected for the heater shall be fire-retardant and shall not emit noxious or toxic fumes at temperatures up to 50 ‘F above the high temperature shut-off temperature (see 3.6.5.4).

3.4 Design and construction. The heater shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation and service.

3.4.1 Exterior openings. All heater exterior openings (exhaust, air inlet, air outlet, control panels, etc.) that may be in the open position during heater operation shall be designed to include means for closing such openings when the heater is not in operation, so that the elements specified in 3.6.7.2 will not collect in quantities that will prevent satisfactory operation of the heater.

3.4.2 Safety. All rotating, reciprocating, or moving parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes and other lines which may be subject to high temperatures and exposed to contact by personnel, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhausts or discharges from the equipment shall be directed so that they do not endanger personnel. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazards. The concept of system safety as defined in MIL-STD-882 shall be followed to assure adequate safety engineering criteria are integrated into the life cycle of the system, i.e., hazards are identified, categorized and eliminated or controlled as early in the life cycle as possible.

3.4.3 Tiedown and hoisting provisions. Four rings with a minimum inside diameter of one inch, shall be provided for tiedown of the heater within an aircraft. Tiedown ring strength shall be consistent with requirements of 3.4.4. The rings shall be of adequate design to permit hoisting of the heater when using a sling. The heater and its parts shall not be deformed or otherwise damaged by use of the sling for hoisting the heater.

3.4.4 Air transportability. The heater shall be constructed to meet the requirements of MIL-A-8421.

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3.4.5 Protection of moving parts. The heater shall be constructed to prevent large foreign particles from entering the blower or blowers while the heater is in operation.

3.4.6 Controls and instruments. The design of the respective assemblies shall be such that all controls and instruments are located in a compact, composite manner. All instruments and controls for operation of the heater, except those for the power package system (see 3.6.6) and the air system (see 3.6.2), shall be readily visible and accessible from the top during normal operation of the heater, by opening a small cover without the necessity of opening a main access door.

3.4.7 Locking devices. All screws, pins, bolts, etc., shall be furnished with locking devices, safety wire, or self-locking nuts. Cotter pins and lock-washers will be acceptable. Locking devices and common parts not manufactured from corrosion-resistant metals shall be cadmium-, zinc-, or nickel-plated in accordance with MIL-STD-171. Locking devices shall prevent the heater hardware from working loose in service. Those subject to removal or adjustment shall not be swagged, peened, staked or otherwise permanently installed.

3.4.8 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided. That mounting shall allow the item to be installed only in its proper operating position.

3.4.9 Common parts. Maximum practicable use shall be made of interchangeable hardware and fastening devices, and of a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common parts.

3.4.10 Fasteners. All steel fasteners shall be plated with cadmium, zinc, or nickel in accordance with MIL-STD-171 to resist atmospheric corrosion under the environmental conditions specified herein. Sheet metal or self-tapping screws shall not be used. Threads shall conform to FED-STD-H28.

3.4.11 Detachable fittings. Disconnect plugs and receptacles in the electrical system, and readily detachable- and attachable-type fittings in mechanical components, shall be used where practicable to permit easy component removal and replacement.

3.4.12 Bearing lubrication. Bearings in the motor, fuel pump and fan shall be sealed and permanently lubricated for operation in an ambient temperature range of -65 to 250 °F. If compatible with all other requirements, a fuel pump design utilizing fuel for lubrication is an acceptable alternative for the fuel pump bearing lubrication.

3.4.13 Maintainability. The heater design and construction shall provide for or accomplish the following:

a. Insure that normal adjustments and repairs can be readily accomplished by operating personnel in the field.

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b. Minimum number of components and part types, designs, and configurations, consistent with the reliability and performance specified herein.
c. Minimum amount of training and time necessary, for assembly, disassembly, location of trouble sources, and maintenance, including servicing. Where practical parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.
d. Permit adjustments, servicing, replacement of parts and components, and other maintenance and servicing with minimum disturbance to other equipment parts or components. The igniter shall be serviceable without removal of any component other than an access panel. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. If engineering reasoning or data determines that physical or visual interference between items for maintenance or servicing cannot be avoided, the items predicted to require the most maintenance shall, where practical, be located for best accessibility.
e. Permit maintenance with general-purpose tools and equipment normally available commercially. Any necessity for use of special-purpose tools and equipment shall be subject to procuring activity approval.
f. Minimum number of tools required for maintenance by design practices such as reducing variety of bolthead sizes to the practical minimum.

3.4.13.1 Overtorquing. Intricate locking devices, controls, and threaded fastenings that can be easily overtorqued by personnel lacking feel through thick gloves or numbness shall be avoided where possible.

3.4.13.2 Quick-disconnect access. Covers or plates that must be removed for component adjustments shall be equipped with substantial quick-disconnect fasteners.

3.4.13.3 Operation clearance. To the maximum practicable extent, the heater shall be designed to make operation, servicing, and maintenance at low ambient temperatures convenient for personnel wearing heavy gloves or mittens and handicapped by bulky clothing and footgear.

3.4.14 Human factors engineering. The heater shall conform to Human Factors Engineering Design Criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, paragraphs 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for Maintainability) and 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

3.5 Performance. The heater shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Operation in temperatures ranging from -65 to 70 °F and exposure to temperatures ranging from -65 to +160 °F.
b. Exposure to and operation in rain as specified in MIL-STD-810, method 506.2, procedure I, with 40 miles per hour (mph) wind and rainfall rate of 2 to 5 inches per hour.

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c. Operation in pressures to 23.98 inches Hg (6,000 feet maximum operating altitude and exposure to 16.9 inches Hg (15,000 feet-storage altitude).
d. Exposure to humidity as specified in MLL-STD-810, method 507.2, procedure III (aggravated exposure), figure 507.2-3, 10 cycles.
e. Exposure to salt-fog as specified in MIL-STD-810, method 509.2, procedure I, 48 hours exposure.

3.5.1 Rated output at -65 °F ambient. At sea level altitude pressure, the heater shall be capable of delivering (in the form of uncontaminated heated air) a heat output of not less than 400,000 Btu/hr at a maximum discharge temperature of 280 °F against a static resistance of not less than 1 inch water column measured at the heater discharge. The airflow rate shall be a minimum of 81 pounds per minute at the discharge of the heat exchanger when operated under the above conditions.

3.5.2 Average heater outlet air temperature. The heater shall maintain a 280 “F average heater outlet air temperature, without reducing blower speed, when the heater is operating in an ambient temperature of -65 °F and is discharging through one set of straightly extended air ducts as specified in 3.6.8.5 or a test duct imposing an equivalent backpressure.

3.5.3 Reduced output at -65 °F ambient. At sea level altitude pressure, the heater shall be capable of delivering (in the form of uncontaminated heated air) a heat output of not more than 125,000 Btu/hr at a discharge temperature Of 150 °F (reduced airflow) against a static resistance of not less than 1 inch water column measured at the heater discharge.

3.5.4 Output at 70 °F ambient. At sea level altitude pressure, the heater shall be capable of delivering (in the form of uncontaminated heated air) a heat output of not less than 180,000 Btu/hr at a discharge temperature of 280 °F against a static resistance of not less than 1 inch water column measured at the heater discharge. With all other conditions unchanged and the discharge temperature adjusted to 150 °F, the heat output shall not exceed 100,000 Btu/hr.

3.5.5 Airflow output at 70 °F ambient. At sea level altitude pressure and an ambient temperature of 70 °F, the heater shall be capable of delivering a minimum of 60 pounds per minute of heated uncontaminated air at a discharge temperature of 280 °F against a static resistance of not less than one inch water column measured at the heater discharge.

3.5.6 Fuels. The class 1, 2, and 3 heaters shall be designed to operate on all aircraft gasolines conforming to MIL-G-5572, automotive gasolines conforming to MIL-G-3056 and automotive gasolines conforming to VV-G-1690. The class 4 heater shall be designed to start and operate on diesel fuel conforming to VV-F-800 and turbine fuel conforming to MIL-T-5624 and MIL-T-83133 under the following conditions:

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Ambient Temp	Specification	Military Symbol
Above +20 °F	VV-F-800	DF-2
Above -25 °F	VV-F-800	DF-1
Above -60 °F	VV-F-800	DF-A
Alternate Fuels		
Above -65 °F	MIL-T-5624	JP-4
Above -50 °F	MIL-T-5624	JP-5
Above -40 °F	MIL-T-83133	JP-8

3.5.7 Tilted operation. The heater shall be capable of functioning satisfactorily when its normal operating plane is at an angle of 8-1/2 degrees in any direction from true horizontal.

3.5.8 Durability. The heater shall be capable of operating for a minimum of 1,000 hours without failure of any major component, as defined in 3.1.1 except for accessories. Cleaning and replacement of parts shall not exceed the limitations specified in 3.5.8.1 and 3.5.8.2.

3.5.8.1 Cleaning. Cleaning of the fuel nozzle and the heater igniter shall be permitted not more than eight times in 1,000 operating hours. Cleaning of lead scale and combustion products from the heat exchanger, burner, and combustion chamber shall not be required more frequently than each 500 operating hours.

3.5.8.2 Parts replacement. For all classes of heaters, replacement of the fuel nozzle and the heater igniter shall be permitted not more than twice in 1,000 operating hours. For class 1 heaters only, replacement of the engine shall be permitted not more than once in 1,000 operating hours. Replacement of any part other than as indicated herein shall not be permitted before completion of 1,000 operating hours.

3.5.9 Reliability. The heater shall have a Mean-Time-Between-Failure (MTBF) of at least 250 hours for class 1 and class 4, 480 hours for class 2, and 185 hours for class 3, when tested as specified in 4.5.2.15.

3.5.10 Maintenance ratio. Class 1, class 3, and class 4 heaters shall demonstrate a maintenance ratio of not more than 0.05, and class 2 heaters shall demonstrate a maintenance ratio of not more than 0.02 when tested as specified in 4.5.2.16. Maintenance ratio is defined as the ratio of the total active maintenance manhours required (scheduled and unscheduled) to the total operating time. Manhours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.5.11 Maintenance downtime. The heater shall be designed and constructed to permit maintenance by military maintenance technicians within the downtimes listed in 3.5.11.1 and 3.5.11.2. Not more than one technician shall be required for any task.

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3. 5.11.1 Corrective maintenance downtime. The mean corrective maintenance downtime for all levels of maintenance below depot maintenance shall be not more than 1 hour with a related maximum corrective maintenance downtime of not more than 3 hours.

3.5.11.2 Maximum preventive maintenance downtime. Maximum preventive maintenance downtime, excluding depot maintenance and major overhaul, shall be not more than 1 hour at the 0.9 percentile point.

3.5.12 Noise limits. The noise produced by the heater shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, paragraphs 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.9. The provision of MIL-STD-1474, paragraphs 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, paragraph 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-i474, paragraph 4.3, except the sign will state “HEARING PROTECTION REQUIRED WITHIN (specify) FEET.” The sign shall be readable at the distance indicated on the sign.

3.5.13 Electromagnetic interference. The electromagnetic interference emission requirements of the heaters shall conform to MIL-STD-461A, Notice 4. Specific subtests of MIL-STD-461A shall be RE02 and CE03 for the class 2 heater and RE05 for class 1, class 3, and class 4 heaters (see 6.5).

3.5.14 Carbon monoxide (CO). The carbon monoxide content of the heating or ventilating air discharged from the heater shall not exceed 35 parts per million (PPM).

3.5.15 Efficiency. With the heater controls set for a discharge temperature of 280 °F and maximum airflow, the heater shall operate with an indicated efficiency of not less than 70 percent and with a measured efficiency of not less than 65 percent. The indicated efficiency shall be based on the flue-gas analysis, and the measured efficiency shall be based on the measured output.

3.6 Details of components.

3.6.1 Combustion system. The combustion system shall provide for the functions of the burner, combustion chamber, and heat exchanger. The combustion system shall be designed for continuous, safe and efficient combustion of the fuel specified in 3.5.6 for all specified conditions of heat output, ambient temperature and barometric pressure. The combustion system shall not fail from exposure to any condition encountered during testing specified in this specification. The system shall be designed to permit disassembly and cleaning as allowed in 3.5.8.1. When exposed to conditions of severe explosion that can occur with fuel-air mixtures, the system shall not rupture or become distorted enough to affect heater performance.

3.6.1.1 Burner. The burner shall provide a means of introducing and mixing fuel and combustion air for ignition and initiation of the combustion process. The burner shall operate without flameout upon rapid manual adjustment of the temperature selector valve for increasing or decreasing outlet air temperature.

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3.6.1.2 Combustion chamber. The combustion chamber shall be a separable, removable chamber or a part of the heat exchanger and shall serve to continue mixing and completion of the combustion process. An observation port shall be provided for the purpose of observing the ignition point and the combustion process. The observation port shall not leak combustion gases to the heated air, and shall be so located that it does not become clogged or clouded by the products of combustion.

3.6.1.3 Heat exchanger. The heat exchanger shall ‘provide for the necessary heat transfer between the combustion gases and the ventilating air to be heated. The heat exchanger shall be gastight to prevent any of the combustion products from entering the ventilating airstream. The heat exchanger shall be designed for maximum efficiency consistent with maintaining the discharge temperature of the exhaust gases above the dewpoint in the heater stack. The exhaust gases shall be discharged vertically in such a manner as to prevent them from being drawn into the ventilating airstream. The exhaust passage shall incorporate a spark arrestor. The heat exchanger shall have a drain that will drain condensate overboard without collecting on the heater surface during operating conditions ranging from -65 to +70 °F.

3.6.2 Air system. The ventilating air and combustion air blower (or blowers) shall provide ventilating air in quantities as specified in 3.5.1 and 3.5.5 and combustion air in quantities necessary for satisfactory combustion of fuels specified in 3.5.6 at all operating condition specified herein. The air system shall include a control means for varying the ventilation airflow rate for compatibility with the performance requirements of 3.5.1 through 3.5.5. The blower or blowers shall be fastened in such a manner as to permit removal for service and maintenance. Exposed blower inlets shall incorporate a louvered sheet metal safety screen to protect operating personnel during normal maintenance and operation of the heater. An arrow indicating the direction of rotation of the fan shall be affixed to the top of the heater fan housing. The inlet safety screen shall prevent any particles larger than a 1/2 inch cube from entering the blower.

3.6.2.1 Air distribution. The ventilating air shall be distributed or mixed so that the heated air temperature at any point leaving the heater shall not vary more than 30 °F from the average temperature. The temperature measuring points shall be located in a traverse 8 to 10 inches from the heater outlet.

3.6.3 Fuel system. The fuel system shall include the filter, pump, lines and fittings, gaskets and seals, safety drains, and tanks.

3.6.3.1 Fuel filter. A fuel filter shall be provided for removal of any foreign particles and suspended ice crystals from the fuel to avoid adverse effect upon heater operation. The filter shall be of the cleanable, reusable type with a metal bowl. The filter shall be mounted in an upright position, and located for quick accessibility for removal, cleaning, and replacement of the filter element and the filter assembly.

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3.6.3.2 Fuel pump. A fuel pump shall be provided to furnish fuel to the burner. The pump shall be compatible with the fuels specified in 3.5.6 and shall supply fuel in quantities and at pressures necessary to meet performance requirements specified herein. The pump shall be mechanically driven by the prime mover.

3.6.3.3 Fuel lines. All fuel lines in the heater shall comply with the fire-proof requirements of SAE AS 1055, class B. Where flexibility is required in these lines, flexible hose assemblies shall be used. The fuel lines shall be compatible with applicable fuels (see 3.5.6) and environments (see 3.5). All fuel lines shall be easily disconnectable and removable for servicing. Where necessary, fuel shutoff valves shall be used to prevent continuous spillage of fuel while disconnecting fuel lines. All high pressure fuel lines within the combustion air plenum shall have provisions to drain any fuel leakage overboard. Such provisions shall avoid traps, and other fuel holding pockets. Test ports (tees) shall be installed as necessary (minimum of three) at various points in the fuel system for the purpose of trouble shooting without the necessity of disconnecting the fuel lines. All threads shall conform to FED-STD-H28. The fuel lines, fittings, and controls shall be sufficiently isolated from the combustion chamber to prevent their being adversely affected by flame, radiant heat, or backfire.

3.6.3.4 Gaskets and seals. All gaskets, synthetic rubber seals, etc., shall be suitable for use with aromatic fuels and shall be satisfactory for use at temperatures encountered within the overheating limits of the heater assembly, and at the cold ambient temperatures specified.

3.6.3.5 Safety fuel drain outlets. Safety fuel drain outlets shall be provided to prevent fuel accumulation anywhere within or on the heater cabinet. This includes fuel accumulation in the combustion chamber, heat exchanger, the floor of the engine compartment, or anywhere else within or on heater cabinet components.

3.6.3.6 Fuel tank. The fuel tank shall be of sufficient capacity to operate the heater and prime mover, when applicable, at maximum rated heat output for a minimum period of 7 hours. The fuel tank shall be an easily removable component of the heater and shall not form any part of the structural frame of the heater. Unless otherwise specified herein, the fuel tank shall conform to MIL-T-46786, type I, class 1, except that the vent test, slosh test and fire test shall not be required.

3.6.3.6.1 Fuel tank fill opening, cap and strainer. The fuel filler neck, cap, and strainer shall conform to the requirements of table II, item no. 2 of MIL-T-46786.

3.6.3.6.2 Drain valve. The fuel tank drain opening shall be equipped with a replaceable, leakproof valve which will allow drainage of accumulated water without the use of tools.

3.6.3.6.3 Fuel indicator. The fuel tank shall be equipped with a non-electrical fuel-level type indicator, which shall be located so that it is readily visible during refueling.

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3.6.3.6.4 Fuel pickup. The fuel tank shall be equipped with a suction tube type of fuel pickup as required by MIL-T-46786.

3.6.3.6.5 Construction. The fuel tank material selection shall be in accordance with table I of MIL-T-46786. Joints and intersections shall be soldered whenever practicable to minimize damage to plating or other protective coatings which may result from unnecessary welding.

3.6.4 Ignition system. A spark ignition system shall be provided for ignition of fuel in the burner/combustion chamber. Ignition shall be sustained during operation of the heater.

3.6.4.1 Igniter. The igniter shall be capable of functioning for 250 hours without service. The igniter shall be fastened to the combustion chamber in a manner to facilitate easy removal for maintenance.

3.6.5 Control system. All controls and instruments required for operation of the heater, except those for the prime mover system and the air system (see 3.6.6 and 3.6.2), shall be located together on a compact control panel. The panel shall be readily visible and accessible by opening a small cover on the top of the heater, and opening of the heater's main access covers shall not be necessary. On class 4 heaters the control panel shall be equipped with a light and switch to aid visibility under poor lighting conditions. When necessary, the control panel or individual controls and instruments shall be shock mounted. The control system shall include a manual outlet air temperature adjustment, temperature sensing device and indicator, high temperature limit device, automatic fuel control device, and provision for starting and stopping combustion. Except for the temperature sensing and indicating device, one or more of the above control functions may be incorporated into one control.

3.6.5.1 Outlet temperature control. A control shall be provided to automatically maintain any desired average discharge air temperature within the range from 150 to 280 °F. The control shall be manually adjustable and equipped with knob or lever to permit selection of the desired temperature. The temperature control shall operate in conjunction with the automatic fuel control device, and be designed to maintain the required temperature range without adjustment to other valves and without permitting burner cutoff due to overheat or flameout over an operating ambient temperature range from +70 to -65 °F. The control shall indicate motion required to increase or decrease the outlet temperature, and shall have limiting stops to prevent selection of outlet temperatures higher than 300 °F.

3.6.5.2 Fuel control device. The automatic fuel control device shall be so related to the manual air temperature adjustument that any desired average outlet temperature between 150 to 280 °F may be automatically maintained within ±10 °F when once established. The device shall be capable of preventing burner flameout during rapid change (up or down) of the air temperature control. The device shall limit fuel flow to prevent external heat exchanger surface temperatures from exceeding 1250 °F during start up at any setting, of the air temperature control.

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3.6.5.3 Temperature indicator. A temperature-sensing device and indicator capable of indicating the average discharge air temperature within ±20 °F of the average discharge temperature shall be provided. The temperature indicator shall be calibrated for the operating range of the heater assembly. The portion of the indicator dial face below the 140 °F mark shall be green, and above the 280 °F mark shall be red. The portion between 140 and 280 °F shall be white. The temperature indicator shall be independent of the operating controls. The temperature-sensing device shall be located close to the heater air discharge and shall be shielded from radiation effects. The device shall respond to the discharge air temperature and transmit this response to the temperature indicator.

3.6.5.4 High-temperature limit device. A high-temperature limit device shall be provided to prevent overheating under any condition of airflow. The device shall be automatically actuated, but manually reset. The high-temperature limit device shall actuate to cease combustion before the average outlet air temperature can exceed 350 °F. The device shall not cause cessation of combustion during normal operation of the heater, such as temperature overshoot (not exceeding 350 °F) when the outlet temperature control is adjusted.

3.6.5.5 Start-stop control. A control shall be provided for manually starting and stopping the burner combustion and fuel flow. The control shall enable the unit to operate without combustion for applications using ventilating air, with no heat.

3.6.5.6 Fail-safe fuel shutoff. The fuel control system shall include a means for fail-safe shutoff of fuel flow to the burner. This feature shall be an inherent part of the fuel control system or a separate, readily accessible manual shutoff valve shall be located in the supply line to the burner nozzle.

3.6.6 Power package systems. A power package system, as specified in 3.6.6.1, 3.6.6.2, 3.6.6.3, or 3.6.6.4, shall be furnished as appropriate for the heater class specified (see 1.2). Each power package shall be a complete subassembly containing all necessary components for installation into the heater assembly and for supporting operation of the heater in conformance with requirements of this specification. Design of the engine exhaust system for all class 1, 3 and 4 heaters shall be fail-safe, such that contamination of the heated air supply cannot result from failure of the exhaust system. Except for class 4 heaters (see 1.2), all power packages shall be readily interchangeable, i.e., electric-to-electric, engine-to-engine, engine-to-electric, and vice versa. The interchangeability design shall make it possible for one man to change the power package within a maximum time of five minutes and without the use of tools. A prototype of each interchangeable power package type shall be available for use in demonstrating interchangeability between types.

3.6.6.1 Class 1 - commercial gasoline engine power package. The commercial gasoline engine power package shall include a gasoline engine, operating controls, and the necessary provisions for mounting the engine and adapting the power package to the other components of the heater.

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3.6.6.1.1 Commercial gasoline engine. For the purpose of selecting the engine, the heater shall be classified as a class 11 end item in accordance with MIL-STD-1410. The engine, including all systems, components} and accessories> shall conform to MIL-STD-1410, except as otherwise specified herein. Data describing the engine selection criteria shall be furnished to enable Government evaluation of the proposed engine prior to award of a contract or order (see 6.8). The engine shall be 4-cycle and air cooled. The engine shall be furnished as a complete self-contained, power package including: hand throttle control, fail-safe governor, manual starting system and engine stop switch. The engine shall be capable of operating under all conditions specified herein when using fuels and lubricants specified herein and in MIL-STD-1410, without exceeding any of the operating limits specified herein and in table I of MIL-STD-1410, or the engine manufacturer's recommended operating limits.

3.6.6.1.2 Engine operating ambient temperatures. The engine shall be so designed that it can be easily started and operated over an ambient temperature range of -65 to +100 °F. For ambient temperatures from -10 to -65 °F, starting operation may be accomplished by providing adequate self-sustained preheating equipment. The engine shall be started and operated within 15 minutes, including preheat time, at ambient temperatures as low as -65 °F.

3.6.6.1.3 Engine air cleaner. The engine shall be provided with a dry type air cleaner and an air cleaner restriction indicator to indicate when the air cleaner element needs replacing. The air cleaner shall not require servicing at intervals of less than 50 hours of operation. If necessary for continuous operation under ice, fog, and snow conditions, air preheat shall be provided at the air filter/carburetor inlet.

3.6.6.1.4 Engine, muffler and spark arrestor. The engine shall be provided with an exhaust muffling and spark arresting means. The muffler and spark arrestor (may be combination muffler-spark arrestor) shall be fabricated of high-temperature, corrosion-resistant material and shall be approved by the U.S. Forest Service Standard 5100-1. The complete exhaust system as installed shall effectively reduce the noise level without developing backpressure in excess of that recommended by the engine manufacturer.

3.6.6.1.5 Engine-heater compatibility. Throughout the range of its operating speed, the engine shall be free of vibrations which may result in damage to the heater or its components.

3.6.6.1.6 Engine preheater. A preheater may be provided for preheating of engine components for starting at low temperatures. Provisions shall be made to prevent spillage of fuel from the preheater over the engine and heater assembly. Excess fuel shall be carried overboard. Preheater fuel controls shall be spring-loaded or otherwise designed so that fuel cannot flow without knowledge of, or action by, the operator.

3.6.6.2 Class 2 - electric motor power package. The electric motor power package shall include an electric motor, operating controls, and the necessary provisions for mounting of the motor and adapting the power package to the other components of the heater.

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3.6.6.2.1 Electric motor. The motor shall operate from 208/220 volt, 60-cycle, 3-phase alternating current. The motor shall conform to NEMA Standard MG-1 and shall be of drip-proof or totally enclosed construction. It shall be furnished with prelubricated and sealed-type ball bearings and shall be entirely suitable for starting and operation throughout the ambient range of -65 to +100 'F. Motor size and speeds shall be left to the discretion of the manufacturer and shall be consistent with the requirements specified herein.

3.6.6.2.2 Power lead cable. The motor shall be equipped with a 20-foot length of 4-conductor power lead cable securely fastened to the motor junction box and provided with l/4-inch individual terminals at the power source end. Provisions for storage of the cable shall be made on the power package, such that the cable can be enclosed within the heater when not in use. The cable shall be resistant to fuels and lubricants and shall withstand low temperatures in accordance with MIL-C-5756.

3.6.6.2.3 Electric motor controls. The electric motor package shall be equipped with an on-off switch, and a speed control if ventilating airflow depends upon motor speed. The controls shall be of adequate rating for the power loads imposed and shall be of outdoor weatherproof construction.

3.6.6.3 Class 3 - military standard gasoline engine power package. The military standard gasoline engine power package shall include the military standard engine as specified in 3.10 with its standard controls, and necessary provisions for mounting the engine and adapting the power package to the other components of the heater. The adaptation of the engine to the heater shall be in accordance with MIL-STD-1226. To assist in starting at extreme low temperatures (-25 to -65 °F), design of the military standard engine power package shall provide access for preheating of the engine with Winterization Torch NSN 4940-00-475-1574. Access shall provide for preheating of the oil pan, flywheel, cylinder head and walls, intake manifold, fuel pumps and fuel filter.

3.6.6.4 Class 4 - diesel engine power package. The diesel engine power package shall include a diesel engine, operating controls, and necessary provisions for mounting the engine and adapting the power package to the other components of the heater.

3.6.6.4.1 Diesel engine. For the purpose of selecting the engine, the heater shall be classified as a class II end item in accordance with MIL-STD-1410. The engine including all systems, components and accessories shall conform to MIL-STD-1410, except as otherwise specified herein. Data describing the engine selection criteria shall be furnished to enable Government evaluation of the proposed engine prior to award of a contract or order (see 6.8). The engine shall be 4-cycle and air-cooled. The engine shall have a 12-volt electric cranking system with MS35000-1 battery, charging generator/alternator and charging receptacles, all in accordance with MIL-STD-1410.

3.6.6.4.2 Diesel engine operating ambient temperatures. The engine shall be so designed that it can be easily started and operated over an ambient temperature range of -65 to +100 °F, and over an ambient pressure range from sea level (0-500 feet) to 6000 ft. For ambient temperatures from -10 to -65 °F,

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starting operation may be accomplished by providing adequate self-sustained preheating equipment. The engine shall be started and operated within 7 minutes, including preheat time, at ambient temperatures as low as -65 °F utilizing JP-4 jet fuel conforming to MIL-T-5624 and as low as -60 °F on DF-A diesel fuel conforming to VV-F-800. The installed battery shall be capable of starting and operating the heater after the temperature has been raised and stabilized at -30 from -65 “F.

3.6.6.4.3 Diesel engine air cleaner. Air cleaner requirements for the diesel engine shall be the same as specified for the commercial gasoline engine in 3.6.6.1.3.

3.6.6.4.4 Diesel engine muffler and spark arrestor. A means of engine muffling and spark arresting shall be provided for the diesel engine. These requirements shall be the same as specified for the commercial gasoline engine in 3.6.6.1.4.

3.6.6.4.5 Diesel engine-heater compatibility. Throughout the range of its operating speed, the diesel engine shall be free of vibrations which may result in damage to the heater or its components.

3.6.6.4.6 Diesel engine preheater. A preheater may be provided for preheating the diesel engine components if necessary for starting at low temperatures. The requirements for the diesel engine preheater shall be the same as for the commercial gasoline engines in 3.6.6.1.6.

3.6.6.4.7 Diesel engine emergency shut-down. An emergency fuel shut-off valve shall be provided on the diesel engine. The valve shall be so located in the fuel system that response shall be within three seconds after actuation.

3.6.7 Housing and chassis assembly. The heater assembly shall be fully enclosed in a lightweight, substantially constructed metal cabinet. The cabinet shall be integrally mounted on a two-wheeled chassis. The wheels and hubs shall comply with MIL-W-8005. The tire size shall be 6.90-9, 6 ply.

3.6.7.1 Accessibility. All working parts of the various sub-assemblies of the heater shall be accessible through convenient doors or removable panels. Controls or control panels shall be accessible through convenient doors. Access doors which could cause injury to the operator or other damage by inadvertent closing shall be provided with safety links or be otherwise locked in the open position. The engine compartment cover shall be hinged to the housing and shall be provided with a locking means to hold the cover in the open position when starting and servicing the engine. The hinge shall be of the continuous type extending the width of the cover.

3.6.7.2 Weather resistance. The cabinets shall resist weather conditions and prevent rain, water, snow, ice, sand, mud, or dust from collecting inside in quantities which would prevent satisfactory operation of the heater.

3.6.7.3 Thermal insulation. The interior of the cabinets shall be insulated or shielded, where necessary, to prevent exterior cabinet surface temperatures from exceeding 140 ‘F at an ambient temperature of 70 °F.

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3.6.7.4 Drainage provisions. The heater cabinet shall have adequate provisions for draining from the compartments any fuel leakage from the fuel lines or controls.

3.6.7.5 Duct and exhaust stack stowage. Secure stowage shall be provided on the heater for the exhaust stack specified in 3.6.8.1 and all of the air ducts and the duct adapter specified in 3.6.8.5.

3.6.7.6 Towing provisions. The heater assembly design shall incorporate pintle and towing provisions so that five heaters can be towed when hooked together one behind the other. The tow bar shall be designed to preclude damage to the heaters during towing operations. A safety device shall be provided to prevent inadvertent disconnection of the tow bar from the pintle. Five heaters hooked one behind the other in line shall be capable of being towed behind a towing vehicle turning on a 15-foot radius for 180 degrees on a level plane with no interference between heaters or damage to the heater cabinet, frame, pintle, or tow bar.

3.6.7.7 Tow bar and parking stand. The heater shall be equipped with either a fixed tow bar and kick parking stand or a combination tow bar/parking stand. The parking stand shall function with the wheels to provide a three point support for the heater in the normal operating position. Heater assembly weight distribution dry or with any amount of fuel in the tank shall provide safe and stable configuration.

3.6.7.8 Tow bar protection. The tow bar shall be designed to withstand damage from a drop to the ground or parking surface from a height equal to the normal-towing position of the tow bar.

3.6.7.9 Springs. The heater shall be equipped with muitileaf, compression coil, torsion coil, air, or equivalent, spring suspensions of sufficient strength to sustain the heater and all associated equipment under all specified operating conditions without evidence of overload or permanent set. Clearance between the springs and spring stops shall be sufficient to prevent frequent bottoming.

3.6.7.10 Parking brakes. The brakes shall lock both wheels so that the wheels will skid and not roll on dry, level, brushed concrete free of loose material when the heater is subjected to a towing force sufficient to move it. Vibration and shocks encountered on applicable operating surface shall not cause the brakes to engage.

3.6.8 Accessories. The accessories, as described herein shall be provided with each heater to complete the assembly as a self-contained item.

3.6.8.1 Exhaust stack. A removable extension exhaust stack not less than 18 inches long shall be provided for each exhaust opening.

3.6.8.2 Discharge air outlet. The heater cabinet shall incorporate a discharge air outlet having a female connection. The outlet shall be fitted with a rigid collar and shall be provided with three pins located 120 degrees apart for making connection to a 12-inch diameter duct conforming to MIL-H-7365.

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3.6.8.3 Radiation shield. If necessary, the discharge air outlet shall incorporate a radiation shield to reduce the detrimental effect of radiated heat from scorching the fabric duct, which is required to withstand temperatures not exceeding 350 ‘F.

3.6.8.4 Handbook compartment. A handbook compartment shall be provided to accommodate storage of operating manuals. When specified (see 3.10), a cotton duct case will be provided as Government- furnished equipment (GFE) in lieu of a handbook compartment. Although only one or the other is required per heater, the handbook compartment or the cotton duct case shall be located within the heater cabinet.

3.6.8.5 Air duct. The heater assembly shall include one complete set of flexible ducts conforming to MIL-H-7365. Unless otherwise specified (see 6.2), the duct color shall be green, color no. 24052. The set of ducts shall include two 12-inch diameter ducts each 15 feet long and three 6-inch diameter ducts each 15 feet long. A duct adapter capable of connecting a 12-inch diameter duct to three 6-inch diameter ducts shall also be supplied.

3.7 Treatment and painting. The portions of the heater normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type F (for ferrous metals) or type G (for non-ferrous metals). Unless otherwise specified (see 6.2), the heater and all parts that are visible externally shall be green no. 383 conforming to MIL-C 46168, except lubrication fittings, tie-down rings, or other plated parts. Internal components need not conform to this color.

3.8 Identification marking. The heater, component parts, subassemblies and accessories shall be marked for identification in accordance with MIL-STD-130. The heater and power package identification plates shall conform to MIL-P-15024/6, type H, style III, size 12. The identification plates shall be affixed to the heaters and power packages as determined by the assigned Government Quality Assurance Representative.

3.8.1 Heater nomenclature. The nomenclature block of the heater identification plate shall include the following information:

“Heater, Duct type, Portable, 400,000 Btu/hr, Type H-1

“Note: Gasoline engine and 208/220 volt/3 phase/60 hertz electric motor power packages or diesel engine power package available for this heater.”

3.8.2 Power package nomenclature. The nomenclature block of the power package identification plate shall include the following information:

“Power package for type H-1 Heater, class (1, 2, 3 or 4 as applicable).”

3.8.3 Part numbering of interchangeable parts. All parts having the same manufacturer’s part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer’s part numbers and changes thereto.

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3.9 Instruction plates. Each heater shall be equipped with instruction plates including warnings, cautions, diagram, and any special or important procedures to be followed in operating and servicing the heater. The instruction plate shall conform to MIL-P-514, type III, composition A, class 2, or composition C with material in accordance with GG-P-455, grade A, class 1, type optional. The plate shall be 0.020 ±.002 inch thick. Unless otherwise specified herein the words “Warning"–and “Caution” shall be in red, matching color number 31136 of FED-STD-595.

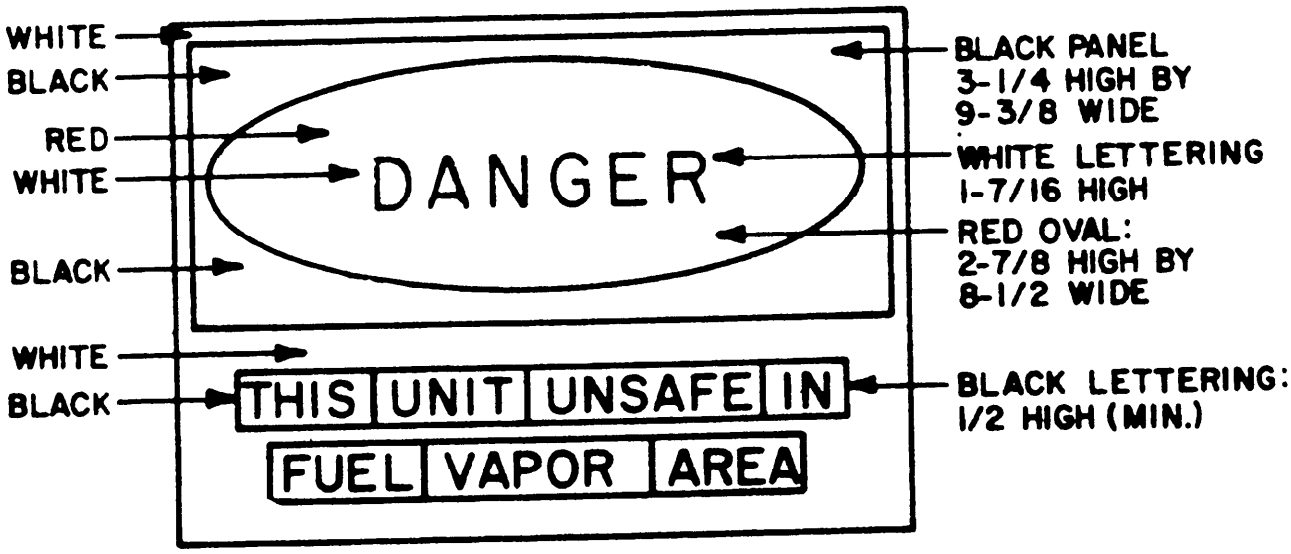
3.9.1 Operating instructions. Brief instructions for starting, operating, and stopping the heater, and notations as to fuels and oils required shall be located on the internal surface of an access door or control panel cover, or on the power package, in such a manner as to be readily visible to the operator when preparing to operate the heater. Reference shall also be made to the technical order or manual number covering operation and service of the heater. Instructions shall be given to run the circulating air blower 2 minutes after shutdown of the burner for protection of the ducts.

3.9.2 Wiring and fuel flow diagram. A combination plate or separate plates depicting wiring and fuel flow diagrams shall be provided for all class 2 and class 4 heaters. Only a fuel flow diagram plate is required for class 1 and class 3 heaters. The required plates shall identify all components and portray the complete operational and functional sequence of the electrical circuit or the fuel system, as appropriate. These plates shall be mounted in a location which is readily visible to personnel servicing these systems.

3.9.3 Warning marking. A danger sign as shown on figure 1 shall be affixed to the side of each heater in a conspicuous location as determined by the Government inspector. The signs shall conform to MIL-P-514, type III, composition A, class 2, or composition C with material in accordance with GG-P-455, grade A, class 1, type optional. The sign shall be 7 inches high, 10 inches wide and 0.020 ±0.002 inch thick. The maximum space available for wording shall be 2-3/4 inches in height and 9-3/8 inches in width.

3.9.4 Other markings. The heater shall be equipped with markings identifying all items that are involved in its operation and handling; such as, controls, tie-down and lifting devices, fuel fill openings, access doors, etc. These markings shall be placed in the most appropriate position suitable for easy viewing as required for safety, operation, and maintenance of the heater.

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NOTES:

1. DIMENSIONS FOR HORIZONTAL PATTERN IN INCHES TOLERANCES: ± 1/8
2. WHITE LINE SEPARATING THE OUTSIDE EDGES OF THE RED OVAL FROM THE ADJACENT EDGE OF THE BLACK PANEL MAY BE USED
3. ALL DIMENSIONS IN INCHES.

FIGURE 1. Danger sign.

X-3690

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3.10 Government-furnished property. When specified (see 3.6.6.3 and 3.6.8.4) the following property in the quantities indicated for each heater will be furnished by the Government (see 6.2), except that item 1 will be furnished only when class 3 heaters are specified:

Item No.	Stock No.	Item Description	
1	2805-00-068-7510	Engine, Gasoline, 1-1/2 HP 4 cycle, Model 1A08-III	1
2	7520-00-559-9618	Cotton Duck Case	1

3.11 Workmanship. Construction and finish of the heater shall give particular attention to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.11.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except where sharp edges or corners are required or where they are not detrimental to safety.

3.11.2 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.11.3 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. Al 1 fasteners shall be correctly torqued and shall have full thread engagement.

3.11.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.11.5 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests prescribed herein. The certification shall be made available for review upon request of the contracting officer or contracting officer's representative. The certification shall provide qualification under tests prescribed by either of the following listed codes for the type of welding operations to be performed and shall show that such qualification is effective as defined the particular code:

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AWS D1.1 Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding." Subject to approval by the Government, contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.11.6 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the heater to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the heater shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

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4.3 First article inspection.

4.3.1 Examination. The first article heaters shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. One first article heater shall be subjected to the tests marked "x" in column 2a of table I, and the other first article heater shall be subjected to the tests marked "X" in column 2b of table I. Failure of any test shall be cause for rejection.

4.3.2.1 Preliminary run-in. Any preliminary run-in of the heater as may be recommended by the contractor, shall be performed by the contractor prior to submission of the heater for first article tests. All data, observations, failures, adjustments, and maintenance obtained or occurring during the run-in shall be included in the first article test report.

TABLEI1. Test schedule.

Test Sequence	Type of test: First Article Test		Type of test: Quality Conformance Test	Test	Quality Conformance Paragraph	Requirement Paragraph
1	2a	2b	3	4	5	6
1	X	-	-	Power package interchangeability	4.5.2.21	3.6.6
2	X	X	X	Leakage	4.5.2.2	3.6.1.3
3	X	X	X	Operational	4.5.2.3	3.5, 3.6.5
4	X	X	-	Performance	4.5.2.4	3.5, 3.6.2, 3.6.5
5	X	X	-	Air contamination	4.5.2.5	3.5.14
6	X	X	-	Controls	4.5.2.6	3.6.5
7	X	-	-	Capacity	4.5.2.7	3.5.1, 3.5.3, 3.5.4, 3.5.5, 3.5.15
8	X	-	-	Tilted-operation	4.5.2.8	3.5.7
9	X	-	-	Noise limits	4.5.2.9	3.5.12
10	X	-	-	Electromagnetic interference	4.5.2.10	3.5.13
11	X	-	-	Altitude	4.5.2.11.1	3.5
12	X	-	-	High temperature	4.5.2.11.2	3.5

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TABLE I. Test schedule. (Cont'd)

Test Sequence	Type of test: First Article Test	Type of test: Quality Conformance Test	Test	Quality Conformance Paragraph	Requirement Paragraph
1	2a	2b	3	4	5	6
13	X	-	-	Low temperature	4.5.2.11.3	3.5, 3.5.1, 3.5.2, 3.5.3, 3.5.15
14	X	-	-	Rain	4.5.2.11.4	3.5
15	X	-	-	Humidity	4.5.2.11.5	3.5
16	X	-	-	Salt fog	4.5.2.11.6	3.5
17	X	-	-	Mobility	4.5.2.12	3.4, 3.6.7.6, 3.6.7.7, 3.6.7.10
18	X	-	-	Tiedown and hoisting	4.5.2.13	3.4.3
*19	-	X	-	Endurance	4.5.2.14	3.5.8, 3.6.8.3
20	X	-	-	Combustion chamber/heat exchanger explosion	4.5.2.18	3.6.1
21	-	X	-	Reliability	4.5.2.15	3.5.9
22	-	X	-	Maintenance ratio	4.5.2.16	3.5.10
23	-	X	-	Maintenance evaluation	4.5.2.17	3.4.13
24	-	X	-	Human factors	4.5.2.19	3.4.14
25	X	-	-	Safety	4.5.2.20	3.4.2

NOTES:
1. Perform tests marked "X".
2. Tests shall be performed in the sequence indicated, or as approved by the contracting officer.
3. For class 4 heaters, test indicated "*" shall be conducted on each first article heater.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each heater shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each heater shall be subjected to the test marked "X" in column 3 of table I. Failure of any test shall be cause for rejection.

4.5 Inspection procedure

4.5.1 Examination. The heater shall be examined for the characteristics

TABLE II. Examination schedule.

Number	Characteristic	Requirement Paragraph
101.	Dimensions not as specified.	3.1.2
102.	Weight not as specified.	3.1.2
103.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
105.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3
106.	Used, rebuilt or remanufactured components pieces, or parts incorporated in the heater.	3.3
107.	Exterior openings not as specified.	3.4.1
108.	Tiedown/hoisting rings not as specified.	3.4.3
109.	Protective screens not providied.	3.4.5
110.	Locking devices not as specified.	3.4.7
111.	Fasteners not as specified.	3.4.10
112.	Detachable fittings not provided.	3.4.11
113.	Bearing lubrication not as specified.	3.4.12
114.	Quick disconnect fasteners not as specified.	3.4.13.2
115.	Combustion system not as specified.	3.6.1 & sub-paragraphs
116.	Blower screen and direction arrow not as specified.	3.6.2
117.	Fuel system not as specified.	3.6.3 & sub-paragraphs
118.	Fuel tank not as specified.	3.6.3.6 & sub-paragraphs

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TABLE II. Examination schedule. (Cont'd)

Number	Characteristic	Requirement Paragraph
119.	Igniter not as specified.	3.6.4.1
120.	Control system not as specified.	3.6.5 & sub-paragraphs
121.	Applicable power package(s) not as specified.	3.6.6 & sub-paragraphs
122.	Housing and chassis not as specified.	3.6.7 & sub-paragraphs
123.	Accessories not as specified.	3.6.8 & sub-paragraphs
124.	Treatment and painting not as specified.	3.7
125.	Identification marking not as specified.	3.8 & sub-paragraphs
126.	Instruction plates not as specified.	3.9, 3.9.1, 3.9.2, 3.9.3
127.	Warning marking not as specified.	3.9.3
128.	Other markings not as specified.	3.9.4
129.	Workmanship not as specified.	3.11 & sub-paragraphs
130.	Paint color not as specified.	3.7

4.5.2 Tests. Tests shall be conducted at prevailing ambient temperatures and utilizing all components and appropriate accessories in place, unless otherwise specified herein. Except as otherwise specified herein, all operating tests shall be conducted with the heater burning a fuel type specified in 3.5.6 or an alternate fuel as approved by the contracting officer. The heater shall be lubricated with appropriate lubricants prior to testing. All instruments shall be calibrated prior to testing. Any test failure attributable to Government-furnished equipment shall not constitute failure of any test specified herein, unless the test failure is determined to be caused by improper application or use of the Government-furnished equipment.

4.5.2.1 Test apparatus and instruments.

4.5.2.1.1 Performance duct. An insulated test duct shall be installed on the heater air outlet for performance determinations other than capacity/airflow tests. The duct shall be a nominal 12 inch diameter and not less than 4 feet long. A thermocouple grid of no less than 20 symmetrically spaced thermocouples shall be located inside of and perpendicular to the duct in a plane at a distance of one foot from the duct discharge end. The thermocouples shall be shielded from the effects of heat exchanger radiation. The duct shall have four static pressure taps located at 90-degree intervals around the circumference in perpendicular plane at a distance of one foot from the duct inlet end.

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4.5.2.1.2 Airflow. Airflow determinations shall be made using a nozzle or orifice type apparatus conforming to ASHRAE Standard 51, AMCA Standard 210 or ASME Code 19.5:4. Other apparatus may be used only when specifically approved by the contracting officer.

4.5.2.1.3 Rotational speed. Rotational speeds shall be measured with a revolution counter and chronometer, a stroboscope and chronometer, a precision instantaneous tachometer, an electronic counter-timer, or any other device which has a demonstrated accuracy of 0.5 percent of the value being measured.

4.5.2.1.4 Fuel consumption. Fuel consumption measurement shall be made by either the volume or weight method. The quantity selected for the volume or weight method shall be such that each reading will cover an elapsed time of at least 15 minutes. Volume measurements shall be corrected for temperature and pressure conditions of the fuel. Flowmeter readings for fuel consumption may be used only when the flowmeter and use thereof has been specifically approved by the procuring activity and the flowmeter has been calibrated for actual test conditions. Regardless of the method used, all expressions of quantities shall be recorded and-converted to a weight basis, unit in pounds per hours.

4.5.2.1.5 Barometric pressure. Barometric pressure shall be measured by means of a mercurial barometer and shall be corrected for the temperature of the scale and the mercury and for the location of the barometer with regard co altitude and latitude. Readings obtained from aneroid barometers will not be acceptable.

4.5.2.1.6 Temperatures. Temperatures shall be measured by means of appropriately located thermocouples and properly calibrated potentiometers. Insofar as practical, thermocouples shall be insulated from contact with other metals and shall be shielded from radiation from the heat exchanger. Laboratory thermometers may be used (shielded if necessary). All temperatures shall be expressed in degrees Fahrenheit.

4.5.2.1.7 Pressures. Static and differential pressures for airflow and performance tests shall be determined by liquid manometers calibrated in inches of water units. Sub-divisions on the indicating scale shall not exceed 0.01 inch of water. Where practicable, other pressures shall be measured by means of liquid manometers. Pressures exceeding the range of conventional, liquid manometers shall be measured by means of properly calibrated Bourdon-tube gauges. Gauges shall be of laboratory type having a dial diameter of not less than 4 inches.

4.5.2.1.8 Exhaust gases. The exhaust gases of the heater shall be determined by analysis with an Orsat apparatus as prescribed by the ASME Power Test Code, Supplement on Instruments and Apparatus, Part 10, Flue Exhaust Gas Analysis. Alternative instruments and procedures for exhaust gas analysis shall be approved by the contracting officer prior to testing.

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4.5.2.2 Leakage. The heat exchanger shall be tested for leaks by subjecting the combustion chamber to an internal air pressure of 5 psig. With the air supply off, the air pressure within the combustion chamber shall exceed 2.5 psig at the end of 5 minutes. The initial leakage test may be conducted prior to installing the heat exchanger into the heater. Air pressure less than 2.5 psig after 5 minutes shall constitute failure of this test.

4.5.2.3 Operational. Each heater shall be started and operated in such a manner as to demonstrate satisfactory operation of all components and to accomplish necessary operational adjustments. Record data to verify proper operation and adjustment. Continue ioeratuib by cycking the burner throught five “on” and “off” cycles. Each cycle shall consist of not less than four minutes of burner “on” time and not less than two minutes of burner “off” time, thus the total “on-off” cycling time shall be not less than 30 minutes. The outlet temperature control shall be adjusted to 280 °F during the “on” portion of the cycle. Normal burner start up and shutdown procedures shall be followed for each cycle. Following the cycling portion of this test, operate the heater and record data to reconfirm proper operation and adjustment of components and controls.

4.5.2.3.1 Basis of failure. One or more of the following shall constitute failure of the operational test:

a. More than 10 percent variation between data recorded before and after the cycling portion of the test.
b. Failure or malfunction of any component, control or accessory.
c. Failure to achieve and sustain burner ignition within one minute of manually activating the burner control.
d. Any evidence of fuel leaks before, during, or after the operational test.

4.5.2.4 Performance. The heater shall be operated at the conditions indicated herein and data as indicated herein shall be recorded to provide a reference base for comparison with other heaters or with subsequent performance tests on the same heater.

4.5.2.4.1 Procedure. The performance test duct per 4.5.2.1.1 shall be installed on the heater and a thermocouple grid shall be attached to the heater return air opening. Provisions shall be included for determination of the exhaust gas temperature, exhaust gas analysis per 4.5.2.1.8, the fuel consumption per 4.5.2.1.4 and the speed of the circulating air fan per 4.5.2.1.3. After all test instrumentation has been installed, adjust the temperature control and airflow to one of the conditions indicated below. Start the heater and allow operation to continue until the outlet air and exhaust gas temperatures have stabilized, such that they do not vary more than 10 °F from their average temperatures. Following stabilization, record three sets of data (see 4.5.2.4.2) at 10-minute intervals, except fuel consumption which shall be measured over a 15-minute period. When all data has been recorded, stop the heater combustion and allow the heater to cool for 5 minutes. Conduct this performance test for each of the following conditions:

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a. Maximum capacity (280 °F) - maximum airflow
b. Minimum capacity (150 °F) - maximum airflow
c. Minimum capacity (150 °F) - minimum airflow
d. Maximum capacity (280 °F) - minimum airflow

4.5.2.4.2 Data. The following data shall be recorded for each test condition of the performance test:

a. Test conditions - capacity/temperature, airflow settings
b. Ambient conditions - temperature -°F, duct static pressure - in H_20
c. Fuel consumption - pounds per hour
d. Inlet air temperature - °F
e. Outlet air temperature - °F
f. Exhaust gas temperature - °F
g. Exhaust gas analysis - percent CO, CO_2 and 0_2
h. Temperature indicator reading - °F
i. Circulating fan speed - rpm
j. Operator observations, adjustments, maintenance

4.5.2.4.3 Basis for failure. One or more of the following shall constitute failure of this test:

a. More than 10 percent variation from like data of other performance tests.
b. Failure of any component or control.
c. Nonconformance to 3.6.5.1 through 3.6.5.5.

4.5.2.5 Air contamination. The heater shall be connected to one of the 12-inch diameter ducts and the three 6-inch diameter ducts shall also be connected utilizing the duct adapter. The heater shall be operated in an ambient temperature between 50 and 70 ±F. Air entering the heater shall not contain more than 15 ppm of carbon monoxide. The heater shall be operated for not less than 30 minutes at each of the following conditions:

a. High heat - high airflow
b. High heat - low airflow
c. Low heat - high airflow
d. Low heat - low airflow

4.5.2.5.1 Data. The following data shall be recorded during the engine starting process and at 10-minute intervals (total of 4 data sets) during each of the 30-minute operating modes. The engine shall be stopped at the end of each operating mode and restarted for the next mode of operation.

a. Temperature entering heater.
b. Temperature leaving heater.
C. Rpm of engine.
d. Mode of operation.
e. Carbon monoxide entering heater.
f. Carbon monoxide leaving heater.

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Temperature measurements shall be made at heater air intake and air discharge openings. Carbon monoxide level shall be measured at the end of the 6-inch flexible ducts using a carbon monoxide monitor accurate to ±5 ppm at the 20 ppm level. The carbon monoxide analyzer shall be calibrated for a zero reading and within ±10 percent of intermediate and maximum allowable levels using test gases with known percentage levels.

4.5.2.5.2 Basis for failure. Nonconformance to 3.5.14 shall constitute failure of this test.

4.5.2.6 Controls.

4.5.2.6.1 Start-stop control. Operation of the start-stop control shall be observed during all other operating tests. The operator shall determine that positioning the control to “start” causes the heater combustion to begin and that positioning the control to “stop” causes combustion to stop and the heater to cool off. Nonconformance with 3.6.5.5 shall constitute failure of this test.

4.5.2.6.2 Outlet temperature control, fuel control and temperature indicator. Proper functioning and performance of the outlet temperature control, the fuel control and the temperature indicator shall be determined from the test specified herein. The test shall be conducted with the performance test duct (see 4.5.2.1.1) installed on the heater. Perform each of the following steps in the sequence listed:

a. Start the heater with the capacity selector set for maximum capacity (280 °F), monitor and record the maximum external heat exchanger surface temperature at the heated air discharge end of the heat exchanger, and observe that the outlet temperature rises and stabilizes without occurrence of flameout or overheat safety shutdown. The external surface temperature of the heat exchanger shall be measured by an appropriate thermocouple attached at a central location on the end surface of the heat exchanger’s primary pass. After the outlet temperature has stabilized, record the conditions of test (ambient temperature, fuel type, location, etc), the outlet air temperature and the temperature indicated by the temperature indicator. Repeat the data recording at 5 minutes intervals until four data sets have been recorded.
b. Instantaneously, reduce the capacity selector setting to minimum capacity (150 °F) and observe that flameout does not occur. After the temperature stabilizes at this setting, record the same data as required in (a) above, four sets at 5-minute intervals.
c. Instantaneously increase the capacity selector setting to maximum capacity (280 °F), monitor and record the maximum external heat exchanger surface temperature, and allow the outlet temperature to stabilize to assure that flameout and overheat safety shutdown do not occur.
d. Set the capacity selector to an intermediate capacity setting, allow the temperature to stabilize, and record data as in (a) above, again 4 sets at 5-minute intervals.

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e. Repeat step (d) above at a second intermediate capacity setting.
f. Conclude the test by stopping the heater combustion and turning the prime mover off after the heater has cooled.

one or more nonconformances under 3.6.5.1 through 3.6.5.5 shall constitute failure of this test.

4.5.2.6.3 High temperature safety. Install the performance test duct (see 4.5.2.1.1) on the heater. The heater discharge airflow shall be partially blocked during heater operation until the high temperature safety control stops the flow of fuel to the combustion chamber. After the air temperature reaches 335 °F the rate of blocking shall be adjusted to limit the rate of average temperature change to 2 “F per minute. The temperature of the discharge air shall be recorded when the safety control deactivates the fuel supply. Nonconformance to 3.3.5 or 3.6.5.4 shall constitute failure of this test. Note: The sensing element of the fuel control device must be deactivated for this test.

4.5.2.7 Capacity. The capacity test shall determine the output capacity, airflow and efficiency characteristics of the heater. The capacity test shall be conducted at normal ambient conditions (70 ±10 °F and sea level to 500 feet elevation) and at low temperature ambient conditions (-65 °F and sea level to 500 feet elevation) (see table I). Each test shall be conducted with the heater discharging heated air against an external static pressure of 1 inch water gauge. Two capacity tests shall be conducted at each ambient condition, one with the capacity selector set for maximum capacity (280 °F outlet air temperature) and the other with the capacity selector set for mimimum capacity (150 °F outlet air temperature).

4.5.2.7.1 Procedure and data. The test apparatus and instruments as specified in 4.5.2.1.2 through 4.5.2.1.8 shall be connected to the heater and set-up as required to collect the data required herein. After all test apparatus and instrumentation has been connected to the heater, operate the heater for not less than 1 hour at test conditions to insure temperature stabilization and uniform operation. Following stabilization, data upon which results will be based shall be recorded every 15 minutes for a total of not less than five complete sets of data. Except for fuel consumption, results shall be based on the average of the five sets of data. Fuel consumption shall be based on the total weight of fuel consumed within the l-hour data-recording period. Each set of data shall consist of not less than the following:

a. Inlet air temperature, °F (read at not less than four locations spaced across the air inlet to the heater).
b. Outlet air temperature, °F (read at not less than four locations spaced across the warm air discharge from the heater and so as not to receive radiant heat).
c. Fuel weight, pounds.
d. Flue-gas temperature, °F (read within 1 inch of where flue-gas analysis sample is taken).

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e. Flue-gas analysis, percent CO_2, O_2, and CO by volume (samples taken from heater exhaust stack at a point 18 inches from the heat exchanger).
f. Static pressure at air discharge, inches of water (read at warm air discharge opening from heater).
g. Static and velocity pressures of heater airflow, inches of water (as required to determine airflow with airflow measuring apparatus being employed).
h. Power package speed, rpm.
i. Barometric pressure, inches of mercury.

4.5.2.7.2 Results. The test data from the capacity test shall be used to determine the following results:

a. Airflow in pounds per minute converted from cfm calculated in accordance with ASHRAE Standard 51, AMCA Standard 210, or ASME Code 19.5:4, as appropriate.
b. Measured output and measured efficiency calculated as follows:

(1) $Q_1 = (cfm)\ (d)\ (t_1 - t_2)\ (0.24)\ (60)$

(2) $Q_2 = W_1 \times HHV$

(3) $E_1 \frac{Q_1}{Q_2} \times 100$

c. Indicated efficiency, E_2, from flue gas losses as determined for all fuels referenced in 3.5.6 by using flue gas temperature and flue gas analysis data in conjunction with no. 2 fuel oil curves, figure 2, page 15.11, ASHRAE 1985 Fundamentals Handbook, as follows:

E_2 = 100 minus percent flue gas losses.

d. Indicated output from measured input and indicated efficiency.

$Q_3 = Q_2 \times 100$

SYMBOLS USED IN CALCULATIONS

Q_1 = measured output, Btu/hr
Q_2 = measured input to unit, Btu/hr
Q_3 = indicated output, Btu/hr
E_1 = measured efficiency, percent
E_2 = Indicated efficiency, percent
d = = air density at temperature measured, lb/cu ft
W_1 = weight of fuel, lb/hr
HHV = highter heating value of fuel, Btu/lb
t_1 = temperature of air leaving unit, °F
t_2 = temperature of air entering unit, °F
cfm = measured airflow, cu. ft./rein

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4.5.2.7.3 Basis for failure. One or more of the following shall constitute failure of this test:

a. Airflow. The heater delivers airflow less than specified in 3.5.1 and 3.5.5.
b. Capacity. The heater has measured capacity not as specified in 3.5.1, 3.5.3 and 3.5.4.
c. Efficiency. The measured or indicated efficiency is less than specified in 3.5.15.

4.5.2.8 Tilted-operation. The complete heater shall be tilted at an angle of 8-1/2 degrees from true horizontal. While tilted at this angle and at an ambient temperature of 60 ±10 °F, the heater shall be started and operated for 30 minutes with controls set for maximum output. This test shall be repeated until the above 30 minutes operation has been completed for each of four 8-1/2 degree tilts at 90-degree increments, radially. Nonconformance to 3.5.7 shall constitute failure of this test.

4.5.2.9 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474 paragraph 5.1.2.1.4 contours shall be taken at not fewer than 12 equal, horizontal arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided at dB(A) level. Failure to comply with MIL-STD-1474 provisions or nonconformance to 3.5.12 shall constitute failure of this test.

4.5.2.10 Electromagnetic interference. The heater shall be tested to determine conformance with 3.5.13. The contractor shall furnish the contracting officer a plan of tests and a report of tests as required by MIL-STD-461. Disapproval of the plan or the report shall constitute failure of this test.

4.5.2.11 Environmental. Each environmental subtest specified herein shall be preceded by and followed by the performance test, condition a, only (see 4.5.2.4) to determine that the functional ability of the heater and its controls has not been impaired by the environmental exposure. A failure of the operational test shall constitute falure of the preceding environmental subtest. After completion of all environmental tests, subject the heater to an air contamination test (see 4.5.2.5). Failure of the air contamination test shall constitute failure of the environmental tests.

4.5.2.11.1 Altitude. The heater shall be subjected to altitude exposure and altitude operating tests. The exposure test shall be in accordance with MIL-STD-810, method 500.2, procedure I. The exposure test shall be a non-operating test and not less than 15,000 feet simulated elevation and at pressure change rates not exceeding 2,000 feet per minute. The operating test shall be conducted at an actual or a simulated elevation of not less than 6,000 feet with an ambient temperature between 60 and 70 °F. The operating test

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shall be conducted in accordance with 4.5.2.4, condition a only, except the operating time shall be extended to record six sets of data at 10-minute intervals. Nonconformance to 3.5 or failure under 4.5.2.4 shall constitute failure of this test.

4.5.2.11.2 High temperature. Subject the heater to a temperature of not less than 160 'F for a period of 4 hours. At the conclusion of the exposure period remove the heater from the test chamber and examine for deterioration, corrosion, or damage. Allow the heater to stabilize at room ambient temperature for a period of not less than 1 hour but not more than 3 hours before conducting post-test operation. Nonconformance to 3.5 or failure under 4.5.2.4 shall constitute failure of this test.

4.5.2.11.3 Low temperature. Subject the heater to a temperature not greater than -65 °F for a continuous period of 12 hours. Following the low temperature exposure and with the temperature remaining at -65 °F, start the heater and operate as required to determine the low temperature capacity performance of the heater. All low temperature operation for class 1, 2, or 3 heaters shall utilize arctic gasoline (MIL-G-3056) or aviation gasolines (MIL-G-5572), and class 4 heaters shall utilize DF-A diesel fuel (VV-F-800) or JP-4 turbine fuel (MIL-T-5624). Aid in low temperature starting of the power package shall be limited to the appropriate equipment permitted by 3.6.6.1.2, 3.6.6.1.6, 3.6.6.3, 3.6.6.4.2, 3.6.6.4.6. Fuel for all low temperature operation shall be subjected to and stabilized at the same temperature as the heater. The low temperature capacity test shall be conducted in accordance with 4.5.2.7. Nonconforamnce to 3.5, 3.5.1, 3.5.2, 3.5.3 or 3.5.15 shall constitute failure of this test.
Note: The exposure and operational portions of the low temperature test may be conducted separately. The exposure portion is to be followed a performance test, condition a only as required by 4.5.2.4, and the operational portion shall be preceded by 8 hours soak at -65 "F prior to starting the heater.

4.5.2.11.4 Rain. Subject the heater to a rain test in accordance with MIL-STD-810, method 506.2, procedure I. The wind velocity shall be 40 mph. The rate of rainfall shall be 2 ±0.5 inches per hour for the first 10 minutes, 5 ±1 inches per hour for the next–5 minutes and 2 ±0.5 inches per hour for the remainder of each 30-minute segment of the test. The heater shall be operated during the test and all covers and doors shall be in their positions for normal operation. Nonconformance to 3.5, failure under 4.5.2.4, burner flameout, or evidence of water accumulation in fuel tanks and reservoirs shall constitute failure of this test.

4.5.2.11.5 Humidity. The heater shall be subjected to the humidity test as specified in MIL-STD-810, method 507.2, procedure III (aggravated exposure), figure 507.2-3. The duration of the test shall be 10 cccles and the heater shall not be operated during exposure, except at the end of the fifth and tenth cycle as indicated in MIL-STD-810. Nonconformance to 3.5 or failure under 4.5.2.4 shall constitute failure of this test.

4.5.2.11.6 Salt fog. The heater shall be subjected to the salt fog test as specified in MIL-STD-810, method 509.2, procedure I. The salt fog exposure

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shall be a continuous 48-hour period. The heater shall not be operated during this test and all covers and doors shall be in their closed positions. Nonconformance to 3.5 or failure under 4.5.2.4 shall constitute failure of this test.

4.5.2.12 Mobility. The complete heater assembly with all accessories in their stowed positions shall be subjected to mobility tests as specified herein. The heater fuel tank shall be full of fuel for all mobility tests except as specified in 4.5.2.12.6. Following completion of each portion of the mobility test the heater shall be examined for misalignment, distortion, fractures or other evidence of structural failure and degradation. Following completion of all portions of the mobility test, conduct a performance test, condition a only, (see 4.5.2.4) and an air contamination test (see 4.5.2.5). Nonconformance to 3.4, 3.6.7.6, 3.6.7.7, or 3.6.7.10; failure under 4.5.2.4 or 4.5.2.5; deformation or failure of any heater part, or any spillage or leakage of fuel shall constitute failure of this test.

4.5.2.12.1 Highway. The heater shall be towed for a distance of 100 miles at a speed of 20 mph over paved highways. Periodically during this test, the equipment shall be turned through a 360-degree turn at its minimum turning radius at a speed of 5 mph.

4.5.2.12.2 Gravel road. The heater shall be towed over graded gravel roads at an average speed of 10 mph (maximum speed 12-1/2 mph) for a distance of 50 miles .

4.5.2.12.3 Rough terrain. The heater shall be towed over rough terrain (unpaved area) at an average speed of 5 mph (maximum speed 7-1/2 mph) for a distance of 5 miles. The rough terrain shall consist of boulder-stem clay or sod and shall be broken by bumps and depressions averaging at least 2 inches above or below the normal surface.

4.5.2.12.4 Multiple tow. A heater and a simulated load representing four other heaters shall be hooked together to simulate the loading of a five heater train. This train shall be towed at an average speed of 10 mph for a distance of 50 miles. A second part of the multiple tow test shall utilize two heaters and a load representing three other heaters to form a simulated five heater train which shall be towed behind a towing vehicle that is guided over a semicircle of 15 foot radius. This turning test shall be repeated until five turns have been completed in the clockwise direction and five turns have been completed in the counterclockwise direction.

4.5.2.12.5 Braking. The heater’s parking brake shall be applied and the fully loaded heater shall be subjected to a forward towing force of sufficient magnitude and duration to move the heater through a distance of six feet. The heater wheels shall be observed for presence of rotary or sliding motion during the six foot movement. Repeat this test with the towing force in a backwards direction.

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4.5.2.12.6 Parking balance. The heater with its fuel tank empty shall be placed in a normal parked position on its two wheels and tow bar/parking stand. Observe whether the heater assembly rests on the tow bar/parking stand in a stable manner. Repeat this test with a fully filled fuel tank.

4.5.2.12.7 Tow bar. The tow bar, mounted on the heater assembly in its normal towing positon, shall be subjected to a drop from a three foot height to a concrete surface. The drop shall be repeated until a total of five drops have been completed.

4.5.2.13 Tiedown and hoisting. The heater assembly, with the fuel tank full, shall be weighted to provide a total weight 2-1/2 times the gross heater weight. A suitable sling shall be attached to the tiedown/hoisting rings and the weighted heater shall be hoisted by the sling to a height of six inches above ground level. The heater shall remain suspended in this condition for 15 minutes. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.2.14 Endurance. The heater shall be subjected to 1000 hours of endurance operation as specified herein. Class 1, 2, or 3 heaters (see 1.2) shall operate for not less than 1000 hours each on automotive gasoline (VV-G-1690). Class 4 heaters (see 1.2) shall operate for not less than 1000-hours each on diesel fuel grade DF-2 (VV-F-800). The 1000 hour endurance operation for each heater shall be subdivided as indicated below:

a. First period - 250 hours at maximum airflow, with discharge temperature of 280 °F.
b. Second period - 250 hours at minimum airflow, with discharge temperature of 280 °F.
c. Third period - 250 hours at maximum airflow, with discharge temperature of 150 °F.
d. Fourth period - 250 hours at minimum airflow, with discharge temperature of 150 °F.

4.5.2.14.1 Procedure. The endurance test shall be conducted in an ambient temperature environment of less than 70 °F (see 6.9). Operation during endurance testing shall be manually controlled, and shall be continuous except as necessary to perform required scheduled maintenance on the heater and the power package. The periods of continuous operation may be altered as necessary to coincide with the work schedules of test personnel. Endurance instrumentation shall be available for maintaining records as required by 4.5.2.14.2. The endurance test for each heater shall be preceded and followed by a leakage test (see 4.5.2.2), a performance test (see 4.5.2.4), an air contamination test (see 4.5.2.5) and a controls test (see 4.5.2.6). The performance test and the controls test shall be repeated at 250-hour intervals during the endurance test. During the second 250-hour endurance period (minimum airflow, 280 °F discharge temperature), the appropriate air duct shall be connected to the heater discharge opening and observed occasionally for indications of heat deterioration.

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4.5.2.14.2 Records. The following data shall be recorded for each heater at the beginning of the test and every 60 hours thereafter, ±10 hours:

a. Fuel rate, pounds per hour.
b. Flue gas temperature, °F.
c. Flue gas analysis, percent by volume of $C0_2$, 02 and CO.
d. Ambient air temperature, °F.
e. Discharge air temperature, °F.

Records shall be maintained for all maintenance actions, operator adjustments, downtime, repairs, etc. The records shall include the time to perform each maintenance action, parts replaced, accumulated operating time, and a full explanation of each service or repair action.

4.5.2.14.3 Failure criteria. One or more of the following shall constitute failure of this test:

a. Nonconformance to 3.5.8.
b. Failure under 4.5.2.2, 4.5.2.4, 4.5.2.5, or 4.5.2.6.
c. Nonconformance to 3.6.8.3.
d. More than ten percent variance in recorded endurance data, except when the cause of variance is attributable to allowable changes in the test environment or allowable adjustments by test personnel.

4.5.2.15 Reliability. The reliability test shall be conducted in accordance with 4.5.2.14, but for the duration specified below. Endurance test time and failures shall be applied toward the reliability test. Data from more than one heater may be included to accumulate total reliability test time. The cycle in 4.5.2.14 shall be repeated until the total reliability test time is accumulated. All failures shall be scored in accordance with 4.5.2.15.1. The test time and allowable number of failures shall be as follows:

Class 1 and 4 heaters:	1250 hours, 6 or less failures
Class 2 heaters:	1500 hours, 4 or less failures
Class 3 heaters:	1100 hours, 8 or less failures

4.5.2.15.1 Failure definition. A failure for purposes of the reliability test is defined as any malfunction which causes or my cause:

a. Damage to the system by continued use.
b. A personal safety hazard.
co Inability to commence operation.
d. Cessation of operation.
e. Degradation of performance beyond specified limits.

Simultaneous related malfunctions are considered as one failure.

4.5.2.16 Maintenance ratio. The maintenance ratio shall be determined from data collected during the reliability test (see 4.5.2.15), Nonconformance to 3.5.10 shall constitute failure of this test.

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4.5.2.17 Maintenance evaluation. Compliance with qualitative maintainability requirements shall be demonstrated by inspection of examination of product. Compliance with quantitative maintenance downtime requirements shall be demonstrated by the task performances specified below. The active maintenance downtimes, the tools and equipment required, and any obstructions or conditions hindering task performance or presenting potential hazard to personnel or to parts shall be observed and recorded. Administrative and supply time, such as completing records or obtaining parts need not count as active downtime. Maintenance task data from other maintenance and teardown inspections done in connection with other tests specified herein may be applied to the maintainability demonstration if adequate observations are made and recorded. Signs of excessive wear or other indications of potential failure shall be recorded where observed and the information supplied to the design activity for study and recommendation of appropriate modifications to be reflected in items to be delivered. Such recommendations shall be subject to procuring activity approval before being implemented and successful repeat of other tests specified herein may be required where the test result could be influenced by the modification. It shall be demonstrated that the heater is in satisfactory operating condition after the maintenance tasks have been done.

4.5.2.17.1 Corrective maintenance. Compliance with the corrective maintenance downtime requirements shall be demonstrated by a maintenance demonstration procedure, maintenance task selection, and maintenance task performance in accordance with MIL-STD-471 including appendices A and B, method 1, test plan pair Al plus B1. The tasks selected and done shall be related co the recommended or to the stated and assumed maintenance concept.

4.5.2.17.2 Preventive maintenance. Compliance with the quantitative preventive maintenance downtime requirements shall be demonstrated in accordance with MIL-STD-471, appendix B, method 6. Each task need not be done more than once, but the resulting times shall be weighed in calculations to account for the expected frequency of each task in the reference period selected. For maximum preventive maintenance downtime demonstration, less than 0.1 (10 percent) of the task times observed shall require greater downtime than the quantitative requirement. Preventive maintenance task shall include servicing and any calibration required.

4.5.2.17.3 Failure criteria. Nonconformance to 3.4.13 shall constitute failure of the maintenance evaluation.

4.5.2.18 Combustion chamber/heat exchanger explosion. A new combustion chamber/heat exchanger assembly shall be subjected to explosion tests as specified herein. The assembly shall have been subjected to a leakage test (see 4.5.2.2), but no other tests shall have been conducted prior to the explosion test. This test shall be conducted once for each combustion chamber/heat exchanger design furnished under the contract, and does not need to be repeated for each heater class when more than one class is specified (see 1.2).

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4.5.2.18.1 Procedure. The explosion teat shall be conducted at an ambient temperature between 60 and 80 °F. A gaseous fuel-air mixture ranging from 0.085 to 0.095 pound of fuel per pound of air shall be introduced into the combustion chamber/heat exchanger assembly until a mixture volume of not less than ten times the volume of the assembly has been passed through the assembly. The mixture shall be ignited. This procedure shall be repeated until five violent explosions have been experienced. Following the five explosions, the assembly shall be examined for rupture and distortion, shall be subjected to a leakage test (see 4.5.2.2), and after installation into a heater, shall be subjected to a performance test (see 4.5.2.4), and an air contamination test (see 4.5.2.5).

4.5.2.18.2 Failure criteria. The following shall constitute failure of this test :

a. Nonconformance to 3.6.1.
b. Failure of the leakage test (see 4.5.2.2).
c. Failure of the performance test (see 4.5.2.4).
d. Failure of the air contamination test (see 4.5.2.5).

4.5.2.19 Human factors. The heater shall be evaluated to determine that the human factors design characteristics have been met. Nonconformance to 3.4.14 shall constitute failure of this test.

4.5.2.20 Safety. Evaluation of safety features shall be accomplished throughtout testing. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.21 Power package interchangeability. When a class 1, 2 or 3 heater is specified (see 1.2), power packages of all three classes shall be interchanged on the heater to demonstrate the adequacy of the interchangeable designs. A list of tools required and the time to change from one power package class to each of the other power package classes shall be recorded. Nonconformance with 3.6.6 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. Examine the first article pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Tests. The first article pack for level A or B shall be subjected to one of the tests specified in 4.7.1.2.1 or 4.7.1.2.2. Failure of the applicable tests shall be cause for rejection.

4.7.1.2.1 Edgewise-drop test. The pack shall be tested in accordance with FED-STD-101, method 5008.1.

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4.7.1.2.2 Pendulum-impact test. The pack shall be tested in accordance with FED-STD-101, method 5012.

4.7.1.2.3 Basis for failure. At the conclusion of the first article pack test, the pack shall be examined. Shifting of contents, visible damage to the contents, or loosening or breaking of anchoring, blocking or bracing within the pack shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

131.	Materials, methods and containers not as specified for level A or B. Each incorrect material, method or container shall be considered one defect.
132.	class 1, 3, and 4 power packages not preserved in accordance with the referenced document and tagged, as specified for level A.
133.	Class 2 power package not preserved in accordance with the referenced document as specified for level A.
134.	Ducts not retracted, wrapped, sealed and stowed as specified for level A.
135.	Duct adapter and exhaust stack not stowed as specified for level A and commercial.
136.	Heater not preserved in accordance with the referenced document, and tagged, as specified for commercial.
137.	Packing not as specified for level A.
138.	Packing not in accordance with the referenced document or shipped mobile as specified for commercial.
139.	Marking, missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 First article pack. Unless otherwise specified (see 6.2) the contractor shall furnish a first article pack for examination and test within the time frame specified, to prove prior to starting production that the applied pre-servation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article heater

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or production heater. If the first article heater is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article heater and the production heater.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Heater. Each heater, with housing and power package in place, shall be preserve in accordance with MIL-P-116, method I and the following specific requirements. Technical publications, when required, shall be preserved in accordance with MIL-P-116, method IC-1 or lC-3.

5.2.1.2 Power packages.

5.2.1.2.1 Class 1, 3, and 4. The power packages fuel systems and related components shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method I, and as specified in 5.4.1.

5.2.1.2.2 Class 2. The power package shall be preserved in accordance with the level A preservation and packaging requirements of MIL-E-16298, using the alternate method as specified for assembled machines.

5.2.1.2.3 Ducts. The flexible ducts shall be retracted, wrapped with barrier material conforming to PPP-B-1055, class C-1 or E-1, sealed with tape conforming to MIL-T-22085, type II, and stowed on the unit in their provided storage locations.

5.2.1.2.4 Duct adapter and exhaust stack. The duct adapter and exhaust stack shall be stowed on the unit in their provided storage locations.

5.2.1.5 Two-wheeled chassis. The two-wheeled chassis shall be disassembled as recommended by the contractor, in a manner to achieve minimum cube, however, any such disassembly shall be limited to removal of wheels, axle, parking stand and other components which are secured to the chassis with bolts and which protrude below the chassis or which otherwise increase cube.

5.2.2 Commercial. Each complete heater, with accessories stowed on the unit as provided for, shall be preserved in accordance with ASTM D 3951 and as specified in 5.4.2.

5.3 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.3.1 Level A. Each complete heater, preserved as specified in 5.2, shall be packed in a close fitting box conforming to PPP-B-601, overseas type, style optional. The heater and chassis components shall be securely blocked, braced, and anchored within the box in a manner to prevent movement or damage in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification.

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5.3.2 Level B. Each complete heater, preserved as specified in 5.2, shall be packed as specified in 5.3.1, except the box shall be domestic type.

5.3.3 Commercial. Unless otherwise specified (see 6.2), each complete heater, preserved as specified in 5.2, shall be packed in a container in accordance with ASTM D 3951. When specified (see 6.2), heaters shall be shipped mobile.

5.4 Fuel.

5.4.1 Level A. A tag certifying that the fuel system has been purged and preserved in accordance-with the applicable requirements of MIL-E-10062 shall be attached to the exterior container.

5.4.2 Commercial. The entire fuel system, except that portion trapped in the lines supplying the burner, shall be completely purged. A tag certifying that this has been accomplished shall be attached to the exterior container or to the heater when shipped mobile.

5.5 Marking.

5.5.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.5.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951, and in addition the weight and cube shall be indicated on the shipping container.

6. NOTES

6.1 Intended use. The type H-1 heater is intended primarily for use in ground heating aircraft engines and fuselages, and secondarily, for use in ground heating maintenance shelters, portable hangers, and other similar enclosures .

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class of heater required (see 1.2).
c. When a first article is required and the time frame for submission of the first article (see 3.2).
d. When a different color is required for the heater and ducts (see 3.6.8.5 and 3.7).

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e. Whether the model 1A08-III gasoline engine furnished with the class 3 heater (see 3.10) is to be furnished by the Government or the contractor.
f. When a cotton duck case is required and whether it is to be furnished by the Government or the contractor (see 3.10).
g. When the welding process shall not be at the option of the contractor (see 3.11.6).
h. When a first article pack is required (see 5.1).
i. Time frame required for submission of first article pack (see 5.1).
j. Degree of preservation and degree of packing required (see 5.2 and 5.3).
k. When each complete heater is not preserved commercially (see 5.3.3).
1. When heaters are required to be shipped mobile (see 5.3.3).

6.3 First article samples. The first article should consist of two preproduction units. The contracting officer shall include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive its requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Any changes or deviations of production heaters from the approved first article heaters during production will be subject to the approval of the contracting officer. Approval of the first article heaters will not relieve the contractor of his obligation to furnish heaters conforming to this specification.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the heaters in accordance with this specification.

6.5 Electromagnetic interference.

a. This heater has been previously procured and qualified to MIL-STD-461A, Notice 1, and MIL-STD-462, Notice 1, as specified in sections 3 and 4 of this specification- Subsequent notices or revisions should not be made applicable to a procurement contract.
b. The contracting officer should include the following in procurement documents:

"(l) Notices or revisions other than those listed below are not applicable:
MIL-STD-461A, Notice 1
MIL-STD-462, Notice 1
(2) Bidders are cautioned against failing to properly take into consideration the radio frequency interference suppression/ electromagnetic interference characteristics requirements for the end items in preparing their bids."

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6.6 Data requirement. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each heater.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of heaters.

6.8 Engine data. The following engine data shall be furnished by bidders with the bid submission to demonstrate that the proposed engine will satisfy the requirements of this specification and MIL-STD-1410. This data shall be furnished in lieu of data specified in 6.3.1 of MIL-STD-1410.

a. Engine description and performance history (see figure 2) or engine manufacturer's certified test data.
b. Engine manufacturer's certification of the engine's suitability for this heater application, including satisfactory operation at the specified environmental conditions and with the specified fuels and lubricants.
c. Description of engine production test procedures with certification of their use for at least 6 months prior to bid.

6.9 Test temperatures. The endurance test may be conducted at ambient temperatures higher than the 70°F specified (see 4.5.2.14.1), however, the contractor is advised to consider that sustained operation at the higher temperatures could accelerate degradation of performance and increase the risk of test failure. Failures resulting from such high temperature operation could place the contract in a delinquent status or unnecessarily obligate the contractor for the additional costs of repeating relevant tests.

6.10 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.10.

6.11 Subject term (key word) listing.

Heater
Heater, diesel fuel
Heater, gasoline burning
Heater, ground shelter

6.12 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army-ME

Review activity:
DLA - CS

Project 4520-A318

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ENGINE DESCRIPTION AND PERFORMANCE HISTORY

1. Engine manufacturer _____

2. Engine Model No. _____ Specification No. _____

3. Horsepower
   Maximum power _____ bhp at _____ Max rpm _____ bmep.

   Maximum intermittent (1 hr) _____ bhp at _____ rpm _____ bmep.

   Net continuous _____ bhp at _____ rpm _____ bmep.

   Horsepower required for end item bhp at _____ rpm _____ bmep.

4. Maximum torque _____ lbs/ft at _____ rpm.

5. Bore _____ in.

6. Stroke _____ in.

7. Total displacement _____ cu in.

8. Major customers for end item application.

Name	Address

9. End item description _____

FIGURE 2

X-3092

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-4607E(ME)	Heater, Engine and Shelter, Ground, Type H-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify): _____
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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